Simulate before you sample: a three-step loop for designing ecological studies

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Abstract

1. Much of what an ecological study can reveal is fixed before any data are collected. Sample size, treatment levels, the range of predictor values and measurement effort together determine which effects can be estimated, and how precisely. Yet these choices are often made by convention, or planned only against a significance threshold, which promotes dichotomous "significant / non-significant" thinking from the outset.

2. Here we propose a practical addition to the research workflow: a *simulation loop*, carried out entirely before data collection. The loop has three steps: (1) translate a conceptual hypothesis into an explicit generative model; (2) simulate and visualise the data that model implies, to scrutinise assumptions about effect sizes, variability and measurement limitations; and (3) iterate design

choices—sample size, treatment levels, predictor range, measurement effort—using criteria focused on estimation quality, especially interval width (precision) and bias, rather than binary detect/not-detect framing. We illustrate the loop with worked linear and non-linear examples and reproducible R code, requiring no specialised statistical machinery beyond base R.

3. The examples show how the loop exposes identifiability limits that no increase in sample size can resolve, how precision can depend as much on *where* you sample as on how much you sample, and how design targets can be set from the precision a scientific or management claim actually requires.

4. Simulating before sampling does not guarantee correct inference; its value is conditional on the assumptions explored. But making those assumptions explicit and inspectable becomes increasingly important as ecological analyses draw on larger datasets and AI-assisted pattern discovery, where deciding what to believe depends on clearly stated assumptions and defensible uncertainty quantification. We intend the loop as a teachable entry point for researchers and students who have never written a generative model.

Keywords: study design, simulation, sample size, precision, statistical power, effect size, generative model, quantitative ecology

1 Introduction

Many of us have seen the same scene in a conference talk, an in-house seminar, or a paper. A speaker presents a compelling story, shows a clean figure with the mean response in a control and a treatment group, and points to the familiar stars of significance, concluding that the treatment “has an effect” ($p < 0.05$)—or, more problematically, that it “has no effect” ($p > 0.05$). The combination of a coherent narrative, an intuitive plot and a thresholded summary can make the conclusion feel settled, even though the same data are often compatible with a wide range of effect sizes. For many listeners and readers, this leaves a quieter set of questions: how much do we actually know from these data, what else are they compatible with, and would the interpretation change under a different but still reasonable analysis?

These concerns are not new. The binary, threshold-based style of statistical reasoning that many of us still use—and are often still taught to use—originated about a century ago (Neyman & Pearson 1933; Gigerenzer 2004) and has been criticised for much of that time, with especially clear calls in recent decades to move beyond “statistical significance” as a scientific standard (e.g. Anderson *et al.* 2000; Amrhein *et al.* 2017; Wasserstein *et al.* 2019; Amrhein *et al.* 2019). Yet the reform has been uneven in ecology and evolutionary biology. Surveys of the published literature find widespread under-powered designs, selective reporting and systematically exaggerated effect sizes (Kimmel *et al.* 2023; Yang *et al.* 2023; Fraser *et al.* 2018), and quantitative training often still centres on the very tests being criticised (Touchon & McCoy 2016).

Most of this reform effort has been directed at how results are analysed and reported. We think an equally important part of the problem sits earlier. By the time data exist, much of what a study can reveal has already been fixed by choices made months before: how many replicates, which treatment levels, how wide a gradient of conditions to span, and how carefully to measure. No subsequent modelling choice can recover information that the sampling design never collected. Worse, a design with poor precision does not simply fail to detect an effect. Conditional on passing a significance filter, such a design reports effects that are systematically too large, and sometimes of the wrong sign (Gelman & Carlin 2014). This is a plausible mechanism behind the exaggeration documented in the ecological literature (Kimmel *et al.* 2023), and it is a property of the design rather than of the analyst. The conventional tool for making these choices, power analysis, inherits the framing we are trying to escape. It asks whether a test will cross a threshold rather than how precisely a quantity will be known, and it is frequently misapplied after the data are in (Hoenig & Heisey 2001). Planning instead for *precision*, that is, for an uncertainty interval narrow enough to support the claim at stake, has been advocated for decades (Goodman & Berlin 1994; Cumming 2013; Rothman & Greenland 2018; Maxwell *et al.* 2008; Lakens 2022). In ecology, however, precision-based planning meets a practical obstacle: closed-form expressions exist only for the simplest designs, while the settings ecologists actually work in, such as saturating responses, nested sampling and non-Gaussian data, have none. What is needed is a way to reach the same answers by simulation.

Here we propose a small but practical addition to the research workflow: a *simulation loop*, carried out before any data are collected. The loop has three steps. First, translate a conceptual hypothesis into an explicit generative model, that is, a formula stating how the data would arise, including both the expected response and its stochastic component. Second, simulate and visualise the data that model implies, and ask whether they look like data the system could plausibly produce. Third, iterate the design, that is, sample size, treatment levels, the range of predictor values and measurement effort, judging each version by the precision it delivers for the parameter that matters rather than by whether a test would clear a threshold. The procedure is a loop because each step can send us back to the previous one: simulated data that look absurd send us back to the model, and a design that cannot deliver useful precision sends us back to the question.

Simulation itself is not new in ecology and evolution, and our contribution is neither the idea nor the machinery but a deliberately narrow and teachable version of them (see DiRenzo *et al.* 2023; Gelman *et al.* 2020; Schad *et al.* 2021; Morris *et al.* 2019, and Table 1 for how the loop relates to other simulation-based approaches). Closest to ours is the four-step workflow of Wolkovich *et al.* (2026), which spans model development through post-fit checking in a Bayesian setting; the loop below stops earlier, asks less, and takes precision rather than calibration as its criterion. We intend it as an entry point for researchers and students who have never written a generative model. Throughout, we provide worked examples with reproducible code: a linear case (Section 3), a non-linear case in which the design question sharpens considerably (Section 4), and two ways in which a precise design can nonetheless mislead (Section 5). Figure 1 places the loop within the conventional research workflow and marks the pitfalls it can help anticipate; Box 1 defines those pitfalls and gives key references.

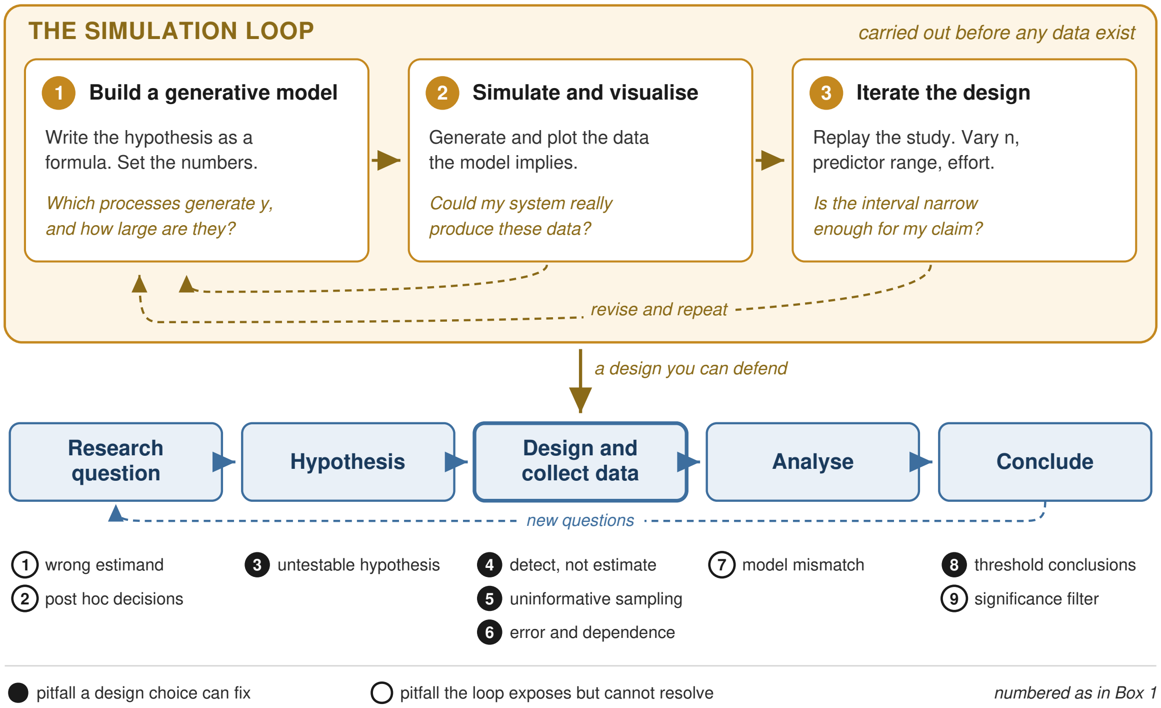

Figure 1: The simulation loop within the conventional research workflow. The lower row shows the familiar sequence from research question to conclusion, with a dashed return path indicating that conclusions raise new questions. The simulation loop (upper panel) is inserted between forming a hypothesis and collecting data, and runs entirely before any observation is made. Its three steps are to translate the hypothesis into an explicit generative model, to simulate and plot the data that model implies, and to iterate the design against a target precision for the parameter of interest. Arrows within the loop run in both directions: simulated data that look implausible send one back to the model, and a design that cannot deliver useful precision sends one back to the question. Numbered markers show the stage at which each pitfall of Box 1 arises. Filled markers (•) are pitfalls a design choice can fix, and which the loop therefore addresses directly; open markers (◦) are those it can expose or quantify but not resolve. The loop is placed where it is because sample size, treatment levels, predictor range and measurement effort together fix what a study can reveal, and no later choice of analysis recovers information the design did not collect.

Box 1: Nine pitfalls along the research workflow

Numbered as in Figure 1, in the order they arise along the workflow. • marks pitfalls a design choice can fix, and which the simulation loop therefore addresses directly; ○ those it can expose or quantify but not resolve.

Framing the question and hypothesis

○ **(1) Estimating something other than the quantity of interest.** An association is reported as though it were a causal effect, because a driver of both the predictor and the response was unmeasured, or because sampling was selective. The estimate may be perfectly good as a description and still answer a different question from the one asked. Refs: Pearl 2019; Grace & Irvine 2020; Arif & MacNeil 2023.

○ **(2) Decisions made after seeing the data.** Hypotheses adjusted to fit the results (*HARKing*; Kerr 1998), analyses continued until a test crosses the threshold (*p-hacking*; Stefan & Schönbrodt 2023), or a single analysis chosen because of how the data happened to look (the *garden of forking paths*; Gelman & Loken 2013). This rarely feels like misconduct, because each individual choice is defensible; what is not defensible is that the choices were available. See also Nickerson 1998.

• **(3) A hypothesis that cannot be wrong.** The claim has no stated direction, no expected magnitude, and no defined alternative, so almost any result can be read as support. Vagueness survives argument but not arithmetic: a hypothesis that cannot be written as a formula with numbers in it was never specific enough to test. Refs: Rajtmajer *et al.* 2022; Gigerenzer 2004; Perezgonzalez 2015.

Designing the study

• **(4) Designing to detect rather than to estimate.** Sample size chosen so that a test is likely to cross a significance threshold, rather than so that the quantity of interest will be known precisely enough to support the claim at stake. The two questions have different answers, and the second is the one a reader needs. Refs: Goodman & Berlin 1994; Rothman & Greenland 2018; Hoenig & Heisey 2001; Halsey *et al.* 2015.

• **(5) Sampling that cannot identify the target.** Predictor values spanning too narrow a range, treatment levels placed where the response is flat, or too few groups to estimate a group-varying effect. Where the observations are taken can matter more than how many there are, and no increase in sample size compensates for a design that never visits the conditions the parameter describes. Refs: Bolker 2019; DiRenzo *et al.* 2023; Section 4.

• **(6) Ignoring how the data actually arise.** Measurement error, imperfect detection, misclassification, and observations that are not independent because they share a plot, a site, or a time. Unlike variation in the system itself, much of this is a design choice: better instruments, repeated readings, or a different allocation of effort. Refs: DiRenzo *et al.* 2023; Popovic *et al.* 2024.

Analysing and reporting

○ **(7) A model that does not match the process.** A saturating response fitted as a straight line, the wrong distribution or link, omitted grouping structure, a transformation that changes what the parameters mean, or extrapolation beyond the range sampled. Such a model can fit well, report a narrow interval, and still be badly wrong about the quantity that matters; regression to the mean is a familiar special case. Refs: DiRenzo *et al.* 2023; Popovic *et al.* 2024; Barnett *et al.* 2005; Wolkovich *et al.* 2021.

• **(8) Conclusions that turn on a threshold.** Results summarised as present or absent rather than as a magnitude with uncertainty, and non-significant findings read as evidence of no effect when they are usually compatible with effects large enough to matter. Reporting the interval instead removes the problem at source. Refs: Amrhein *et al.* 2017, 2019; Wasserstein *et al.* 2019; Muff *et al.* 2022; Singal *et al.* 2017.

○ **(9) The significance filter on the literature.** Because significant results are more likely to be written up and published, the published subset of an imprecise design overstates the effect and sometimes reverses its sign, even when every individual study was analysed correctly. The exaggeration is invisible from inside any one study but computable from the design in advance. Refs: Gelman & Carlin 2014; van Zwet & Cator 2021; Kimmel *et al.* 2023; Yang *et al.* 2023; Ioannidis 2005.

2 The simulation loop

This section sets out the loop in general terms; Sections 3 and 4 work through it on concrete examples. The aim throughout is to make assumptions explicit, to turn a verbal hypothesis into a quantitative expectation, and to find out what a planned design can and cannot support before it is carried out.

2.1 Core idea: an explicit generative model

At the centre of the loop is an explicit generative model, a statement of how the data would come about. A useful starting point separates the systematic component (pattern) from the stochastic component (noise):

$$y_i = f(x_i; \theta) + \varepsilon_i \tag{1}$$

where $f(x_i; \theta)$ is the expected value of y_i given predictors x_i and parameters θ, the signal implied by the hypothesis, and ε_i is the remaining deviation around that expectation. Depending on context, ε_i may represent process variation, observation or measurement error, or both. It is worth stating which, because the two have different design remedies: process variation is a property of the system and must be lived with, whereas measurement error can often be reduced by better instruments, repeated readings or a more careful protocol.

Equation 1 describes how y arises *given* x. For design purposes we must also specify how x itself arises: the treatment allocation, the range of predictor values, the replication structure, the spatial or temporal layout, the measurement schedule. This second half is easy to overlook, and it is where most of the design leverage lives. It is also what distinguishes a design simulation from a model check. The parameters θ are assumptions about nature, which we can only estimate or guess at; the process generating x is a description of what we intend to do, and is entirely under our control. Simulating both makes visible which features of the data are determined by our choices and which are left to chance.

We ask only that the model be explicit enough to simulate from. The more of it that rests on known mechanism, the better: mechanism is what constrains plausible parameter values, so Step 1 is easier and the resulting design more defensible when the model has one.

2.2 The three steps

Step 1: From a verbal hypothesis to a generative model. Models are hypotheses. With every model we fit we are testing an additional set of claims, often without full awareness. Building the model deliberately, from what is already known about the variables and their relationships, constrains this to a defensible set. Which explanatory variables influence y? Is the relationship linear, saturating, or humped? What does the literature suggest about the magnitude of the effect, and about how variable it is between individuals, plots or sites? In our experience a conceptual hypothesis often makes intuitive sense right up to the moment one tries to write it as a formula, at which point it can become clear that it lacks direction, magnitude, or any foundation in prior knowledge. This is uncomfortable, and it is among the most valuable things the loop does. Writing the model down also forces us to state what we actually want to estimate—the estimand—and what values we expect it to take. In ecology this step should anticipate design features such as hierarchy and non-independence, since they change what the data can support.

None of this is quick, and it is worth saying so plainly. Building a model one can simulate from is harder than fitting a model to data, and building one whose output could pass for real data is harder still. The difficulty is structural rather than a matter of practice. A likelihood tells us how to weigh parameter values against data that already exist, but it does not say how those data came to exist (Gelman *et al.* 2020, ch. 5): the sample size, the predictor values, the missingness and the measurement process all sit outside it. A fitted model inherits all of that from the dataset in front of us, and can leave it unstated; a model we simulate from has to supply it. That, and not simply a shortage of coding skill, is the honest answer to an obvious question, namely why a method with such apparent appeal is not already routine.

The effort is not overhead, however. Deciding which processes generate the response, at what magnitude and with what variation, is exactly where a statistical model gets tied to the science it is meant to be about. A model that can only be fitted leaves those commitments implicit, and they are then carried into the results without ever having been examined; a model one can simulate from has been obliged to declare them. The cost of Step 1 is therefore also its return.

Step 2: Simulate and visualise synthetic data. "The greatest value of a picture is when it forces us to notice what we never expected to see" (Tukey 1977). With the model specified, we generate a synthetic dataset from it. The key point is to simulate y *through* the model structure rather than drawing y from some marginal distribution: we generate predictor values under the planned design, compute the expected response via $f(x;\theta)$, and then add stochasticity via ε. Plotting the result shows the hypothesis in the same visual form as a real results figure, with direction, magnitude and consistency together. The question to ask at this point is biological rather than statistical: could this system plausibly produce these data? Implausible scatter, impossible values, or a relationship that looks nothing like what one would expect from the literature are all signs that the assumptions, rather than the simulation, need revisiting.

Step 3: Iterate the design against a precision target. The third step plays the planned study through many times on the computer. This goes beyond a classical power analysis, which targets a significance threshold. We repeatedly simulate data under plausible assumptions, fit the analysis we intend to use, and examine the estimates and their uncertainty. Three questions follow, in order:

1. *Does the analysis recover the values we generated from?* This is a check on the code and on the model implementation, not on the science. If a correct analysis cannot recover parameters from data we generated ourselves under ideal conditions, something is wrong—and we would never have learned it from the empirical fit alone.

2. *How do estimates and uncertainty respond to design choices?* Sample size, the range of predictor values, the number of groups versus replicates per group, and measurement effort are all levers. They differ enormously in cost, and their relative value is rarely obvious in advance.

3. *What would this design do to our conclusions if we interpreted it conventionally?* Simulation makes it straightforward to compute quantities that are otherwise invisible: how often the estimate would carry the wrong sign, and by what factor it would be exaggerated, conditional on passing a significance filter (Gelman & Carlin 2014).

For this to end in a decision it needs a target. The translation is direct: if a claim requires an effect to be known to within $\pm k$, the design target is an uncertainty interval of width about $2k$ (Rothman & Greenland 2018). One refinement matters. A design does not produce a single interval width but a distribution of widths across the studies one might run, so designing to the mean of that distribution leaves roughly half of realised studies short of the target. Designing against a high quantile instead—conventionally the 80th percentile—is known as *assurance* in the clinical trials literature and costs nothing but a change of summary statistic. Section 3 works this through.

Going back and forth between assumptions, simulation and fitting is a playful but powerful way to design a study.

2.3 What the loop is, and what it is not

Simulation is used in several distinct ways in ecology and evolution, and the differences matter because they answer different questions at different points in a project. Table 1 places the simulation loop among them. Our scope is deliberately the narrowest of the five: everything in this paper happens before any data exist, and the criterion throughout is the precision of the estimate we care about. That narrowness has a cost, which Section 5 makes explicit.

Table 1: Five uses of simulation in statistical practice. The simulation loop occupies the earliest and narrowest of these roles. It complements rather than replaces the others; a study designed with the loop would naturally be analysed with a workflow of the kind described by Wolkovich *et al.* (2026) or Gelman *et al.* (2020).

Approach	**Question it answers**	**When applied**	**Criterion**
Classical power analysis	Will my test reject the null hypothesis?	Before data	Power at a threshold
Simulation loop (this paper)	How precisely will I know the quantity I care about?	Before data	Interval width for the estimand
Bayesian and other workflows (Gelman *et al.* 2020; Schad *et al.* 2021)	Is my model, and my inference from it, trustworthy?	Before and during analysis	Calibration, parameter recovery
Validating complex models (DiRenzo *et al.* 2023)	Does this model behave as I think on data like mine?	After model specification	Recovery, identifiability
Simulation studies (Morris *et al.* 2019)	Does this statistical method perform well in general?	Method development	Bias, coverage, RMSE

3 A linear example: plant growth and nitrogen

This first example is deliberately simple and can be reproduced directly; a fully commented R script containing all the code and figures below is available in the Supplement.

3.1 Step 1: from a verbal hypothesis to a generative model

Every study starts with a question. Here: what is the effect of nitrogen fertilisation on plant growth? We plan to grow sunflowers across a gradient of nitrogen concentrations and harvest biomass after one season. The response y is biomass, and the verbal hypothesis is that plants grow more when nitrogen increases.

That sentence is not yet something we can simulate from. To simulate, we must commit to three things it leaves open: the *shape* of the relationship, the *size* of the effect, and its *consistency*. Assuming a positive linear relationship between biomass and nitrogen concentration x gives

$$y_i = a + b\,x_i + \varepsilon_i, \qquad \varepsilon_i \sim \mathrm{N}(0, \sigma) \tag{2}$$

where a is the intercept, b is the slope—the expected biomass increase per unit nitrogen, and our estimand—and σ captures variability around the mean relationship. We now have to say what we think these parameters are:

```
# Our assumptions about the system, stated as numbers
a     <- 30   # intercept: expected biomass (g) at zero nitrogen
b     <-  7   # slope: biomass gain (g) per unit nitrogen; the estimand
sigma <-  5   # residual SD (g): process variation plus measurement error
```

These values are not truth. They are assumptions, made explicit so that they can be argued with, and everything that follows is conditional on them. That is the point rather than a limitation: it moves the debate from the sample size to the reasoning behind it.

3.2 Step 2: simulate the data the model implies

We choose a design and generate data through the model structure: draw predictor values as the design specifies, compute the expected response, then add noise.

```
# Choose a design and simulate one study from the model above
n <- 50                                          # number of plants
x <- rnorm(n, mean = 10, sd = 4)                 # planned nitrogen gradient
y <- a + b * x + rnorm(n, mean = 0, sd = sigma)  # expected response, plus noise
fake <- data.frame(x = x, y = y)
```

In an experiment with fixed treatment levels rather than a gradient, only the line generating `x` changes, for example `x <- rep(c(0, 5, 10, 15, 20), each = 10)`. The rest of the loop is identical. Plotting the result (Fig. 2a) shows the hypothesis in the same visual form as a real results figure. The question to ask is biological rather than statistical: could sunflowers in this system plausibly produce these data? If the scatter looks far too tight, or the biomass values are impossible, then the assumptions are wrong, and it is very much cheaper to discover that now than after a field season. We can also fit the model we intend to use and check that it recovers what we put in (Fig. 2b):

```
# Fit the planned analysis and look at the interval, not the p-value
fit_lm <- lm(y ~ x, data = fake)
confint(fit_lm)["x", ]      # 95% interval for the slope b
```

At this point it is tempting to look at the p-value. We deliberately do not. The quantity we carry forward is the *width* of the interval for b (Fig. 2c). A narrow interval means the study pinned the nitrogen effect down; a wide one means the data are compatible with many different effect sizes, whatever the p-value says. Note also that this is a single study. Run it again and the interval shifts and changes length.

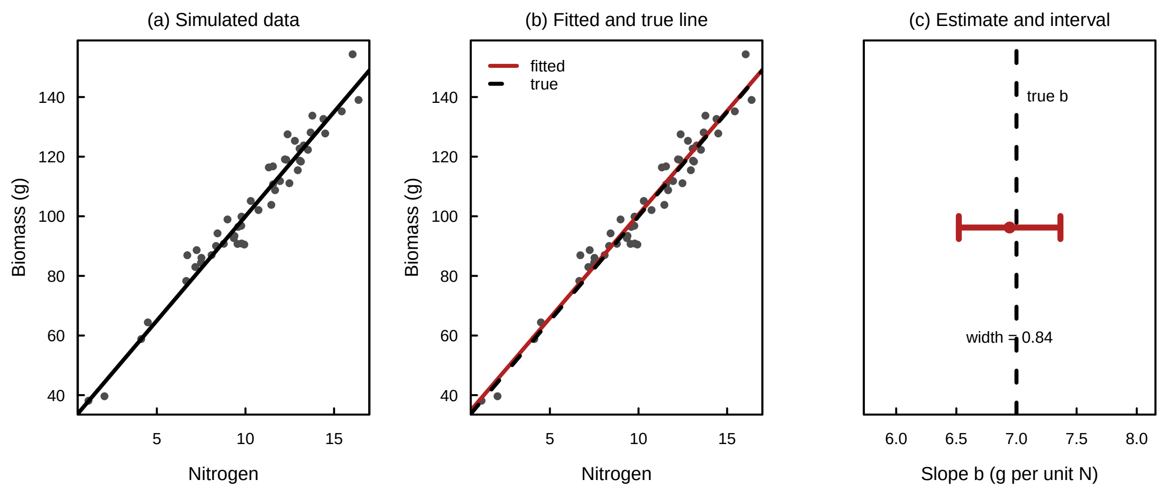

Figure 2: Step 2 of the loop for the linear example, from a single simulated study. (a) Data simulated from Equation 2 with $a = 30$, $b = 7$, $\sigma = 5$ and $n = 50$, shown with the generating line: the hypothesis in visual form, with effect size and scatter together. (b) The fitted line (solid) against the generating line (dashed), a check on the code rather than on the science. (c) The estimated slope (point) with its 95% interval (bar) and the true value (dashed). The width of that bar, printed beside it, is the quantity the rest of the section is about.

3.3 Step 3: how precision responds to design

We now play the planned study through many times. One run generates a dataset, fits the intended model, and records the interval width for b:

```
# One simulated study: generate data, fit the model, return the interval width
one_run <- function(n, a, b, sigma, mean_x = 10, sd_x = 4) {
        x  <- rnorm(n, mean = mean_x, sd = sd_x)
        y  <- a + b * x + rnorm(n, 0, sigma)
        ci <- confint(lm(y ~ x))["x", ]
        unname(ci[2] - ci[1])
}

# Repeat the same study many times to see what this design typically delivers
widths <- replicate(500, one_run(50, a, b, sigma))
```

Repeating this across sample sizes traces a design-precision curve (Fig. 3a). Precision improves with n, but with sharply diminishing returns: interval width falls roughly as $1/\sqrt{n}$, so halving the interval costs four times the sample size. The figure carries two lines because a design does not have a precision but a distribution of precisions (Fig. 3b), and the 80th percentile is the honest summary of what a single realised study is likely to achieve.

It is worth pausing on why we can trust any of this. For an ordinary linear model there is a closed form, $\mathrm{se}(b) = \sigma/(s_x\sqrt{n-1})$, and the simulation reproduces it to three decimal places across the whole range of n (the comparison is printed by the Supplement script). Simulation was not necessary here, and a formula would have done. Checking the machinery where it can be checked is what licenses using it in Sections 4 and 5, where no formula exists.

Sample size is not the only lever, and often not the best one. Holding $n = 50$ fixed and varying only the spread of nitrogen values shrinks the interval substantially (Fig. 3c):

```
# Same n, wider nitrogen gradient: how much does spreading x out buy us?
sd_x_values <- c(1, 2, 4, 6, 8)
sapply(sd_x_values, function(s)
mean(replicate(500, one_run(50, a, b, sigma, sd_x = s))))
```

This lever is essentially free, since it costs no more to fertilise across a wider range, and it is invisible to a power calculation that treats n as the only design variable. Measurement quality is a third lever (Fig. 3d): doubling σ costs roughly a fourfold increase in n to recover the same precision, so a better instrument or a repeated reading may be far cheaper than more plants.

3.4 From a scientific requirement to a sample size

Applying the rule from Section 2: suppose a fertiliser recommendation requires the nitrogen effect to within ± 0.25 g biomass per unit nitrogen. The design target is then an interval of width 0.5, read off the curve at the 80th percentile (Fig. 4).

```
# What sample size meets a target width of 0.5 in 80% of studies?
n_grid <- seq(20, 300, by = 20)
q80    <- sapply(n_grid, function(nn)
quantile(replicate(500, one_run(nn, a, b, sigma)), 0.8))
n_grid[which(q80 <= 0.5)[1]]
```

Under the assumptions above this gives $n \approx 120$. The number is conditional on the assumed σ and predictor range, which is a feature rather than a limitation: it makes the basis of the design auditable. A reviewer who disagrees can argue with σ, a statement about the system, rather than with the sample size, which is merely its consequence.

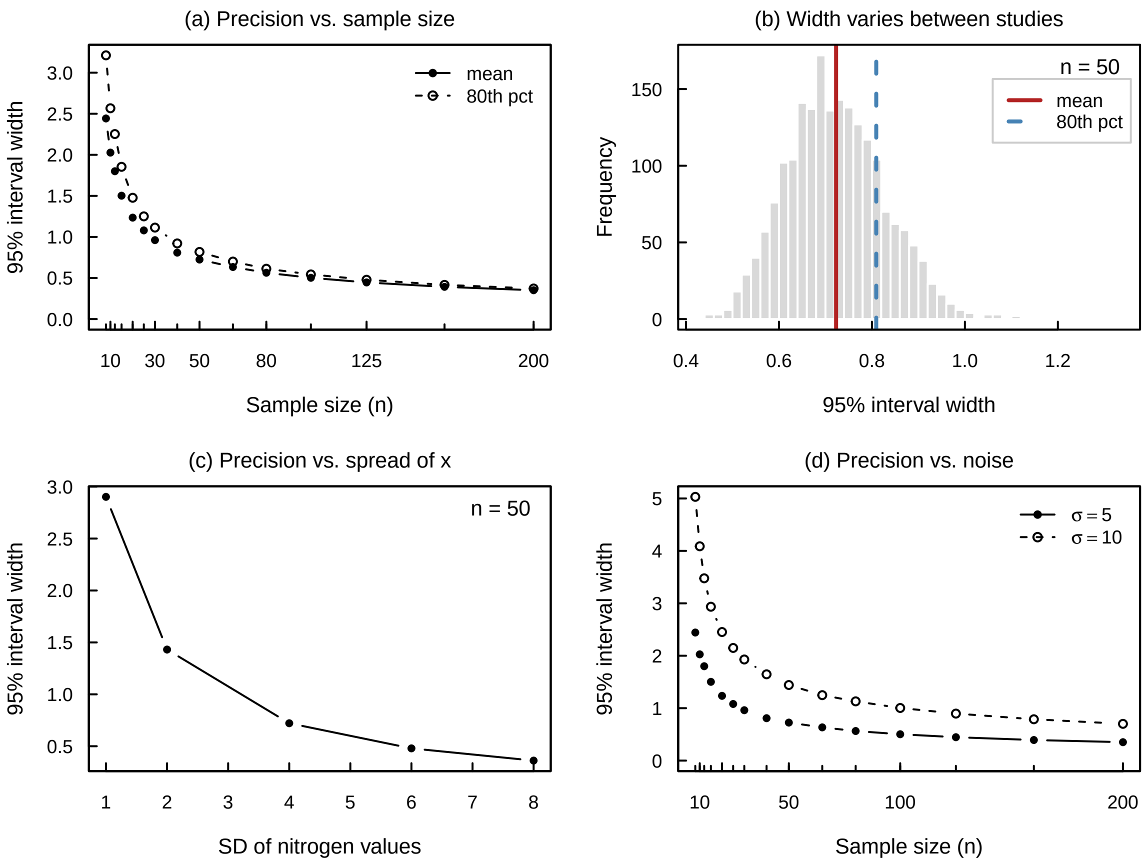

Figure 3: How the precision of the nitrogen effect responds to three design levers, from 500 simulated studies per point. (a) Design-precision curve: mean (filled symbols) and 80th percentile (open symbols) of the 95% interval width against sample size. (b) At a single design ($n = 50$) the width is itself a random variable; the solid line marks its mean and the dashed line its 80th percentile. Designing to the mean leaves about half of realised studies short of the target. (c) At fixed $n = 50$, spreading the predictor values more widely improves precision markedly, at no cost in effort. (d) The curve from (a) against the same design with the residual SD doubled from $\sigma = 5$ to $\sigma = 10$: the noisier design needs roughly four times the sample size for the same precision.

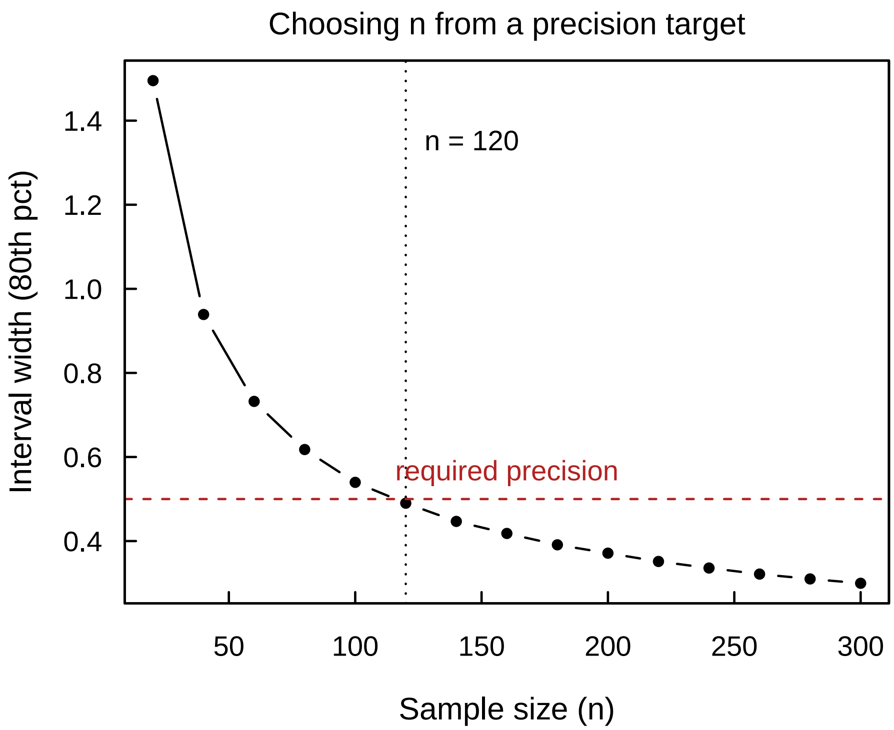

Figure 4: Choosing a sample size from a precision requirement. The curve is the 80th percentile of the 95% interval width for b across 500 simulated studies at each sample size; the horizontal line is the target width of 0.5 implied by a requirement of ± 0.25. The vertical line marks where the two meet, at $n \approx 120$.

3.5 What an imprecise design would do to our conclusions

Everything so far concerns precision. The same machinery also reveals what poor precision does to *interpretation*, which is where the consequences for the published literature lie.

Consider a more typical ecological situation than our flagship example: a small true effect ($b = 0.5$), noisy measurement ($\sigma = 15$) and a modest sample ($n = 20$). Simulating this design and applying a conventional significance filter gives a power of about 9%. That much a power analysis would have told us. What it would not have told us is the rest. Conditional on reaching $p < 0.05$, this design reports an effect roughly four times too large, and carries the wrong sign about 7% of the time (Gelman & Carlin 2014). The distribution of estimates that would be published barely overlaps the truth (Fig. 5). The design therefore does not merely fail to detect the effect; it manufactures misleading results. This cannot be diagnosed after the fact from a single dataset, because the exaggeration is invisible from inside the study that produced it. It is, however, entirely visible beforehand, from the design alone.

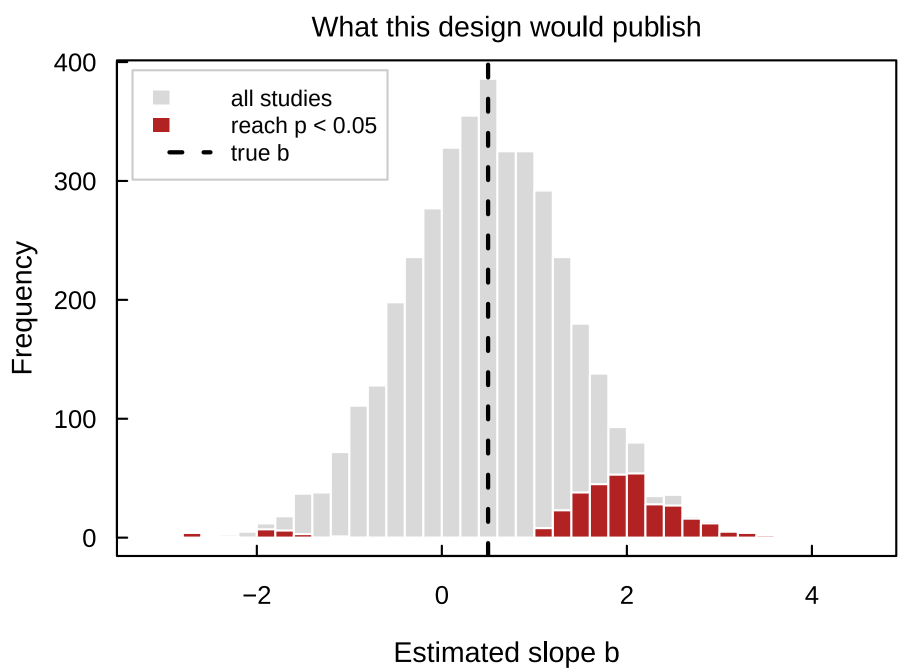

Figure 5: What an imprecise design would publish. Slope estimates from 4000 simulated studies with $b = 0.5$, $\sigma = 15$ and $n = 20$ (light), and the subset reaching $p < 0.05$, which is the part most likely to be reported (dark). The dashed line is the true value. Conditional on significance the design exaggerates the effect roughly fourfold and returns the wrong sign about 7% of the time, so the distribution that would be published barely overlaps the truth.

4 A non-linear example: a saturating response to nitrogen

Many ecological relationships are non-linear: responses saturate, show thresholds, or peak at intermediate conditions. Simulation earns its place here, for two reasons. There is no closed-form expression for the precision of a non-linear parameter, so the analytic shortcut used in Section 3 is unavailable. And the same data can often be explained by quite different parameter combinations, which makes what you can learn depend heavily on where you sample rather than simply on how much.

4.1 Step 1: a saturating model

We keep the same question, but now assume that growth increases with nitrogen and then levels off as other resources become limiting. A standard choice is the Michaelis–Menten function:

$$y_i = \frac{V_{\max}\, x_i}{K + x_i} + \varepsilon_i, \qquad \varepsilon_i \sim \mathrm{N}(0, \sigma) \tag{3}$$

where $V_{\max}$ is the asymptotic biomass, the value approached as nitrogen becomes unlimiting, and K is the half-saturation constant, the nitrogen value at which expected biomass is $V_{\max}/2$. Unlike the slope in Section 3, these two parameters answer different biological questions. $V_{\max}$ describes the ceiling; K describes how quickly that ceiling is approached. A study can pin down one without the other, or neither, and which of those happens is largely a matter of design.

```r
# Assumed parameter values for the saturating model
Vmax  <- 120  # asymptotic biomass (g)
K     <-   8  # half-saturation constant: nitrogen at which y = Vmax / 2
sigma <-  10  # residual SD (g)

# The mean function
mu_mm <- function(x, Vmax, K) Vmax * x / (K + x)
```

Equation 3 forces $y = 0$ at $x = 0$, which is unrealistic if the soil already contains nitrogen. Adding a baseline term is straightforward and makes the identifiability problem below somewhat worse rather than better; we keep the simpler form for clarity.

4.2 Step 2: two candidate designs, same sample size

The key design choice in a non-linear model is the range of x. We simulate two designs with identical sample sizes: one sampling nitrogen only in a low to moderate range, and one extending far enough to approach the asymptote.

```
# Two designs, same n, differing only in how far the nitrogen gradient extends
n <- 50
x_narrow <- runif(n, 0, 12)   # Design A: stops well short of saturation
x_wide   <- runif(n, 0, 40)   # Design B: extends past the bend

fake_narrow <- data.frame(x = x_narrow,
y = mu_mm(x_narrow, Vmax, K) + rnorm(n, 0, sigma))
fake_wide   <- data.frame(x = x_wide,
y = mu_mm(x_wide,   Vmax, K) + rnorm(n, 0, sigma))
```

Plotting the two datasets shows the problem immediately (Fig. 6). Over the narrow range the saturating curve is nearly a straight line. The data contain almost no information about the asymptote, because the design never went anywhere near it. Same n, same model, same noise, but Design A cannot see the thing the model is about.

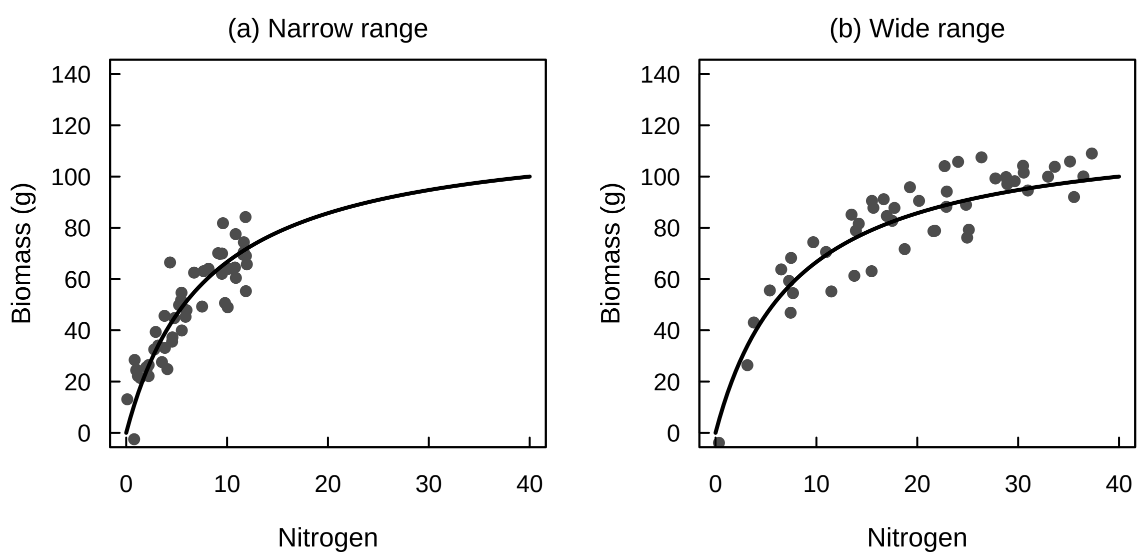

Figure 6: Two designs with the same sample size ($n = 50$) under Equation 3, with $V_{\max} = 120$, $K = 8$ and $\sigma = 10$. Both panels are drawn on the same axes and show the generating curve (solid). (a) A narrow nitrogen range, $x \in [0, 12]$, gives data that look almost linear and carry little information about the asymptote. (b) A wider range, $x \in [0, 40]$, reaches the bend and reveals the saturation.

4.3 Step 3: fitting, convergence, and the cloud of estimates

We fit the planned model with non-linear least squares. Because `nls` can fail to converge, it is tempting to set `warnOnly = TRUE` and move on. We recommend against it: that option returns non-converged fits as though they were estimates, and quietly contaminates everything computed downstream. Recording convergence status instead turns a nuisance into a design diagnostic.

```
# Fit the saturating model; return NULL rather than a non-converged fit
fit_mm <- function(dat) {
	fit <- try(nls(y ~ Vmax * x / (K + x), data = dat,
	start = list(Vmax = max(dat$y), K = median(dat$x)),
	control = nls.control(maxiter = 200)), silent = TRUE)
	if (inherits(fit, "try-error")) NULL else fit
}
```

Under Design A the fit fails to converge in XX% of simulated studies, against YY% under Design B. That difference is a result in its own right: it says the narrow design is asking the data for something they cannot supply. A caution follows immediately. Discarding failed runs before summarising the rest is itself a selection step, of the kind this paper warns against elsewhere, since the surviving estimates are the subset that happened to be informative. How many runs were dropped should be reported alongside the estimates that remain.

Repeating the whole study many times under each design and plotting the resulting parameter estimates against one another makes the underlying problem visible (Fig. 7). Under Design A the estimates smear along a ridge: many combinations of K and $V_{\max}$ produce almost identical curves over $x \in [0, 12]$, so the data cannot tell them apart. A large $V_{\max}$ paired with a large K looks much the same, over that range, as a smaller $V_{\max}$ paired with a smaller K; plotting the curves those estimates imply shows them lying on top of one another wherever the data are, and fanning apart only beyond the sampled range (Fig. 7c). Under Design B the cloud collapses around the true values. This is a design problem rather than an analysis problem. The wide design does not analyse better; it simply visits the conditions under which the asymptote is expressed.

4.4 More data, or better placed data?

The design question can now be put directly. Given a fixed budget, is it better to collect more observations or to spread them over a wider range of nitrogen?

```
# Compare the spread of K estimates across sample sizes and nitrogen ranges
ns <- c(20, 50, 100, 200)
sapply(ns, function(nn) IQR(replicate_K(nn, x_max = 12)))  # narrow
sapply(ns, function(nn) IQR(replicate_K(nn, x_max = 40)))  # wide
```

Increasing the sample size under the narrow design buys less than simply extending the nitrogen

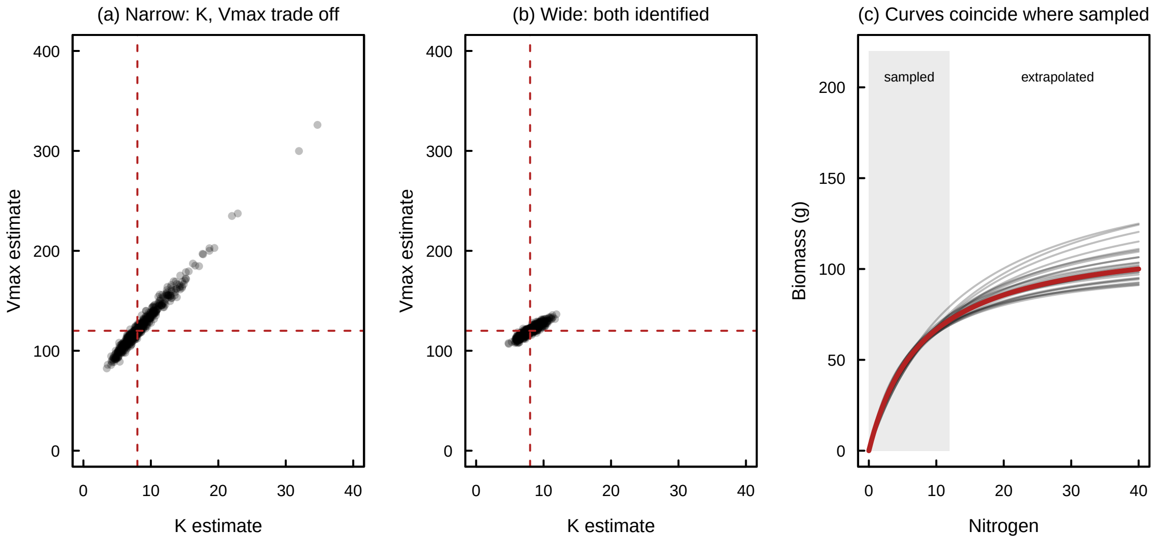

Figure 7: Why the narrow design cannot identify the saturating model. Estimates come from 500 simulated studies under each design at $n = 50$. (a) Under the narrow range the estimates of K and $V_{\max}$ smear along a ridge and neither parameter is determined; dashed lines mark the true values. (b) Under the wide range the cloud collapses onto them. (c) The reason. Each faint curve is the fit from one narrow-design study. Within the sampled region (shaded) they are indistinguishable from one another and from the generating curve (heavy line), and they separate only outside it, where no data were collected. Non-converged fits are excluded throughout, and their frequency is given in the text.

gradient (Fig. 8), and over the range we examined the wide design at the smallest sample size beats the narrow design at the largest. If a project can afford one change, the simulation says which one to make, and it says so before the money is spent. This is the central practical claim of the paper in its sharpest form: identifiability and precision can depend as much on where you sample as on how many observations you collect.

So far the analysis model has always matched the model that generated the data, and the design has always been experimental. Section 5 relaxes each assumption in turn, and shows that neither failure is visible in the precision of the result.

5 When precision is not enough

Precision was the criterion in Sections 3 and 4. This section sets the bound. A 95% interval is a statement about sampling variability under an assumed model. It says nothing about whether that model has the right structure, and nothing about whether the parameter it surrounds is the one the scientific question was about. We demonstrate each failure in turn. In both, the interval is narrow,

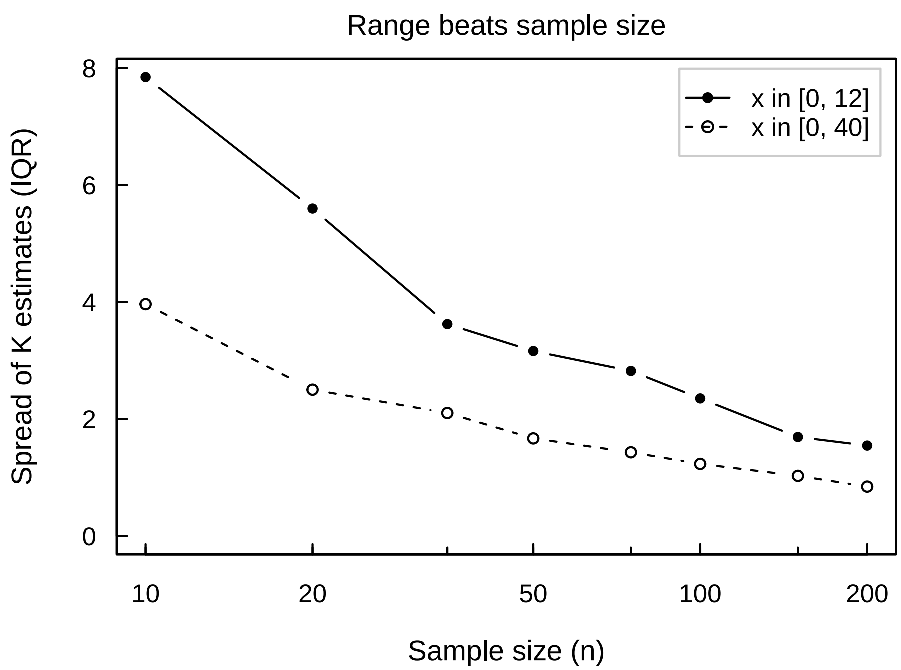

Figure 8: Interquartile range of the estimates of K against sample size on a log scale, under the narrow and wide nitrogen ranges, from 500 simulated studies at each point. Widening the gradient does more for the precision of K than any affordable increase in sample size does within the narrow range: over the range shown, the wide design at the smallest sample size is already better than the narrow design at the largest.

the diagnostics are clean, and the answer is wrong.

5.1 A model with the wrong shape

We fit an ordinary straight line to data generated by the saturating process of Section 4, using the narrow design in which the curve looks almost linear anyway.

```
# Fit a straight line to data generated by the saturating model
fit_wrong <- lm(y ~ x, data = fake_narrow)
summary(fit_wrong)$r.squared                      # looks like a good fit
confint(fit_wrong)["x", ]                         # a tight interval
predict(fit_wrong, newdata = data.frame(x = 40))  # extrapolate to x = 40
mu_mm(40, Vmax, K)                                # what the truth would be
```

Every conventional diagnostic looks reassuring. The straight line explains about three-quarters of the variance ($R^2 = 0.75$), the slope is estimated to within a couple of per cent, and nothing in the output flags a problem. The predicted biomass at $x = 40$ is about 234 g. The true value is 100 g (Fig. 9a). Collecting more data would tighten that interval without moving it, because the interval tightens around a value that was never the target.

Goodness of fit gives no warning either. The saturating model that actually generated these data reaches $R^2 = 0.79$ on the same observations, and that is a ceiling rather than a fitted value, since the curve was not estimated from the data at all. Over this range the correct shape and a badly wrong one are separated by three points of R^2. A respectable fit statistic is therefore not evidence that the structure is right.

There is a constructive reading. The same simulation that exposes the problem points to the remedy: a design extending past the bend both identifies the parameters (Section 4) and makes the inadequacy of a linear model obvious in the data. Designing for precision and designing to be able to detect one's own mistakes turn out, here, to be the same activity.

5.2 A parameter that is not the one we mean

Now suppose the model has the right shape but we cannot randomise. We survey plots across a landscape, measuring nitrogen where it happens to occur. Soil moisture varies across that landscape and affects both: wetter plots mineralise more nitrogen, and wetter plots also grow more biomass for reasons unrelated to nitrogen.

The step that matters is Step 1, and it is easy to skip. The generative model must describe how we believe the data actually arise, including processes we do not intend to measure. Writing moisture into the simulation is not a claim that we will have moisture data; it is a claim about the world.

```
# An observational design in which moisture drives both nitrogen and biomass
n_obs    <- 500
moisture <- rnorm(n_obs, 0, 1)                              # the confounder
x_obs    <- 10 + 3 * moisture + rnorm(n_obs, 0, 1)          # N tracks moisture
y_obs    <- 30 + 7 * x_obs + 15 * moisture + rnorm(n_obs, 0, 5)

naive <- lm(y_obs ~ x_obs)                       # moisture unmeasured
adj   <- lm(y_obs ~ x_obs + moisture)            # moisture measured
```

The causal effect of nitrogen is 7. The naive analysis returns about 11.5, with a 95% interval of roughly $[11.3, 11.7]$ that excludes the truth entirely; the adjusted analysis recovers 6.9 (Fig. 9b). Nothing in the naive output signals a problem: the residuals are well behaved, the model is correctly specified as a straight line, and the estimate is very precise.

Precision is again the difficulty. The bias does not depend on n, so a larger sample narrows the interval without correcting the estimate. At $n = 10\,000$ the naive interval spans 11.44 to 11.53 (Fig. 9c). A large dataset, analysed with a correctly specified model, yields an extremely confident answer to a question nobody asked.

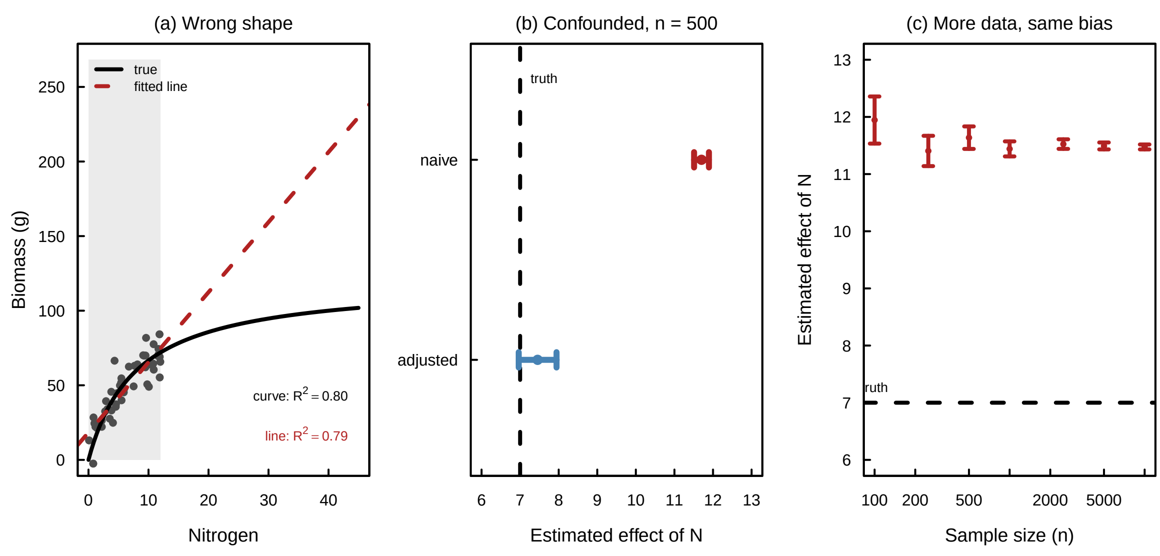

Figure 9: Two ways a precise design can mislead. (a) A straight line (dashed) fitted to data generated by the saturating model (solid) over a narrow nitrogen range (shaded). Each R^2 is labelled with the object it belongs to: the fitted line reaches 0.75 and the true generating curve only 0.79, so goodness of fit barely separates them. Extrapolated to $x = 40$ the line nonetheless overpredicts biomass by more than a factor of two. (b) Estimates of the nitrogen effect at $n = 500$, with 95% intervals, from an analysis omitting soil moisture (naive) and one including it (adjusted); the dashed line is the true causal effect of 7. (c) The naive estimate and its interval as sample size increases from 100 to 10 000: the interval contracts steadily around a value roughly 65% too large.

5.3 What follows for the loop

The simulation loop does not solve either problem. It cannot tell you that your functional form is wrong, and no amount of simulation licenses a causal claim from an observational design; that requires randomisation or an explicit argument about which variables must be measured and adjusted for (Pearl 2019; Grace & Irvine 2020; Arif & MacNeil 2023).

What it offers is earlier warning, and three things follow. It quantifies the damage while it can still be acted on: simulating a confounder one suspects exists takes a few lines and answers a concrete question, and here the answer is 65%, large enough to change what one would do. It disciplines the estimand, because the naive analysis above is a perfectly good estimate of an association and wrong only as an answer to a causal question, a distinction that writing the generative model forces into the open. And it corrects a natural intuition about sample size: it is tempting to treat a large dataset as insurance against error, whereas for this class of problem the larger study is the more confidently wrong one, and the less likely to be questioned.

This is why we treat Step 1 as the step that repays most attention, and the one most often skipped. The precision machinery of Sections 3 and 4 is only as good as the generative model it is conditioned on.

6 Reporting a simulation loop

The loop is only useful to a reader if they can disagree with it productively. That requires reporting the assumptions, not just the resulting design. The checklist below lists what we think a paper or proposal should state; all of it fits comfortably in a short methods paragraph plus a supplement.

1. **The generative model.** Equations for the mean function $f(x;\theta)$ and for the stochastic component ε, on the response scale.

2. **The assumed parameter values,** with their justification: literature, pilot data, or expert judgement labelled as such. State the range explored where a single value could not be defended.

3. **What the error term represents:** process variation, measurement error, or both, and

whether hierarchy or autocorrelation is included.

4. **The design variables explored:** sample size, treatment levels, predictor range, replication structure, measurement effort. Say which were varied and which were held fixed.
5. **The performance criterion,** and the target. We recommend the width of the uncertainty interval for the focal parameter, summarised at a stated quantile rather than as a mean, together with the scientific or management requirement the target derives from.
6. **Failures as well as successes:** the proportion of simulated fits that did not converge, and how they were handled. This is a design diagnostic, not a nuisance.
7. **Reproducibility:** the code, the random seed, and software versions, archived with a persistent identifier.

The same three steps extend without modification to two-group comparisons, generalised responses such as counts and proportions, and hierarchical or repeated-measures designs; only the generative model and the analysis model change. The Supplement works through a nested design in which a fixed budget of plots is allocated across sites in five different ways, a case where no closed-form guidance exists and the differences in precision are substantial.

7 Outlook

The proposal of this paper is narrow. Turn a verbal hypothesis into an explicit generative model, simulate what it implies, and choose a design by the precision it delivers for the parameter that matters. Everything else follows from taking that seriously, including the uncomfortable findings of Section 5.

Advice of this kind is not new, and it is worth asking why it has not been taken up more widely. Part of the answer is the difficulty set out in Section 2, which is intrinsic to the task and which better tools do not remove. The rest was more mundane. Writing a generative model for anything beyond a straight line has required a level of comfort with code that ecological training has mostly reserved for theoreticians, and an empiricist facing a field season has had good reason to skip it. That second barrier is now falling quickly. Code assistants make a first simulation of a nested, non-Gaussian or saturating design something that can be drafted in minutes rather than an afternoon, and revised as easily. The practical case for the loop is therefore stronger today than when much of the same advice was offered a decade ago, not because the argument has improved but because the cost of acting on it has dropped.

What remains expensive is the part that matters most. A model that can be simulated from still has to be defended: which processes generate the data, what magnitudes are plausible, what varies and by how much, and what the study is actually trying to estimate. Those judgements are ecological rather than computational, and nothing about cheaper code supplies them. If anything the balance shifts further towards Step 1, because the steps that used to consume the effort no longer do. The same is true of the design target itself. Choosing an interval width of $2k$ is a statement about how much error is tolerable for the claim at stake, which is a scientific and sometimes a policy judgement rather than a statistical one (Douglas 2000; Steel 2010). Making the generative model explicit does not settle such questions, but it turns them from hidden defaults into inspectable choices (Romeijn 2014; Gelman & Shalizi 2013), which is the most that can be asked of any method (Box 1976).

These considerations become more pressing, not less, as ecological analyses draw on larger datasets and on AI-assisted pattern discovery. Modern methods are very good at finding regularities in high-dimensional data and comparatively poor at telling us which of them to believe (Pichler & Hartig 2023; Lucas 2020). Simulation offers one concrete form of discipline here: a generative model specifies what the data would look like if a proposed mechanism were absent, and a pattern that a method also recovers from such data has not been established by finding it. The confounded example of Section 5 is the more sobering case, because it scales in the wrong direction: the studies that command most confidence are the ones whose bias is hardest to notice. Volume of data is not a substitute for having asked what generated it (Pearl 2019), and reporting standards for uncertainty in ecology remain uneven enough that the difference is easy to miss (Simmonds *et al.* 2024; Lele 2020).

None of this requires a new methodology. It requires doing something ordinary at an unusual time, namely writing down what one expects before finding out, and it costs an afternoon at the point in a project where an afternoon is cheapest. We have found the exercise most valuable when it went badly: when the simulated data looked nothing like the system, when the estimates would not settle however much data we imagined collecting, or when the design we had planned turned out to answer a question adjacent to the one we cared about. Those are useful things to learn in front of a computer, and expensive things to learn in the field.

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